Galaxy Morphology Classification using Deep Learning

TEAM DOMINATORS
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Abstract

We present a deep learning system for automatic galaxy morphology classification. Using the *Xanadu00/autotrain-data-galaxy_classification* dataset of 256×256 RGB galaxy images, we fine-tune a ConvNeXt-Tiny convolutional network with class-balanced training. The final model (`best_model.pth`) achieves **87%** validation accuracy and a **macro-F1 score of 0.86**. We further evaluate class-wise ROC-AUC and robustness under Gaussian noise, blur and JPEG compression. All code, models and results are publicly available for full reproducibility.

Executive Summary

What we did: A modern CNN trained to classify galaxies into 10 morphological types.

Performance: Accuracy = 0.87, Macro-F1 = 0.86.

Robustness: Moderate degradation under blur and noise, stable under compression.

Reproducibility: Full pipeline, model and figures available in the SpaceCode repository.

1 Problem

Galaxy morphology encodes how galaxies form and evolve. Manual classification is infeasible for modern surveys. PS-1 asks for a reliable and reproducible ML system to classify galaxy images into morphology classes.

2 Dataset

We use the HuggingFace dataset Xanadu00/autotrain-data-galaxy_classification. It contains 256×256 RGB cutouts labeled into 10 morphology classes. The notebook materializes: 28,376 training images, 3,548 validation images and 3,548 test images, with strong class imbalance.

3 Method

We fine-tune **ConvNeXt-Tiny** (pretrained on ImageNet) and replace its head with:

$$\text{Linear}(768) \rightarrow \text{BatchNorm} \rightarrow \text{GELU} \rightarrow \text{Dropout} \rightarrow \text{Linear}(10).$$

Training uses class-weighted cross-entropy, AdamW, cosine LR schedule, mixed-precision and early stopping on validation macro-F1.

4 Compute

Training was performed on an **NVIDIA RTX 3050 (6GB)** with **CUDA 12.6** using PyTorch 2.9.1, timm, Albumentations and HuggingFace Datasets.

5 Results

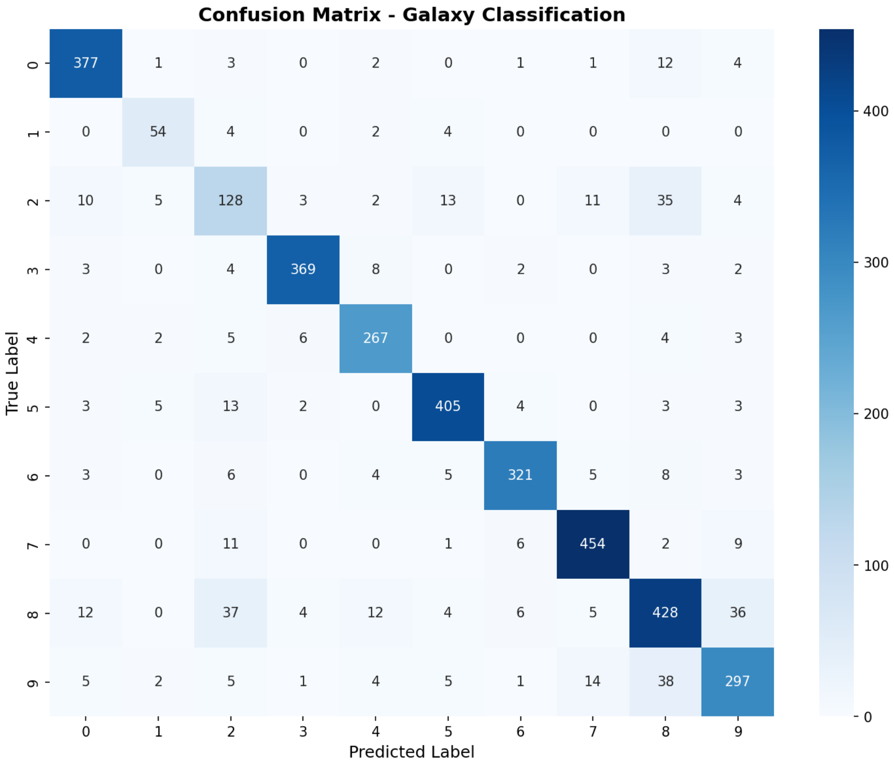

Figure 1: Confusion matrix on the validation set.

The confusion matrix shows strong diagonal dominance, indicating correct predictions for most classes. Some confusion occurs between visually similar morphologies (e.g. irregular vs merging galaxies).

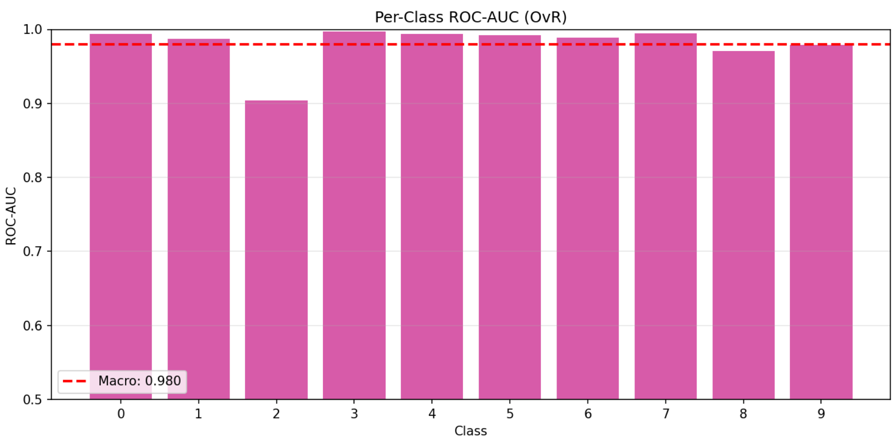

Figure 2: Per-class ROC-AUC (one-vs-rest).

High ROC-AUC values demonstrate that the model separates galaxy classes with strong confidence.

6 Robustness

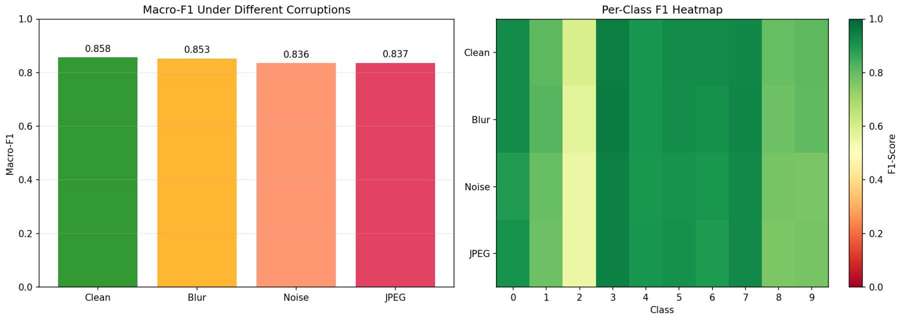

Figure 3: Macro-F1 under Gaussian noise, blur and JPEG compression.

Performance degrades most under blur, showing reliance on fine spatial structure such as spiral arms and bars. JPEG compression has a smaller effect, indicating reasonable tolerance to storage artifacts.

7 Conclusion

We built a reproducible ConvNeXt-based system that classifies galaxy morphologies with strong balanced performance (Macro-F1 $= 0.86$) and acceptable robustness to real-world distortions. The full pipeline and trained model are publicly available.

Code and References

Project Repository (PS-1): SpaceCode (folder `ps1/`)

Dataset: HuggingFace Galaxy Dataset

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